Hyperspectral Unmixing Using Orthogonal Sparse Prior-Based Autoencoder With Hyper-Laplacian Loss and Data-Driven Outlier Detection

Zeyang Dou, Kun Gao, Xiaodian Zhang, Hong Wang, and Junwei Wang

***Abstract*— Hyperspectral unmixing, which estimates end-members and their corresponding abundance fractions simultaneously, is an important task for hyperspectral applications. In this article, we propose a new autoencoder-based hyperspectral unmixing model with three novel components. First, we propose a new sparse prior to abundance maps. The proposed prior, called orthogonal sparse prior (OSP), is based on the observations that different abundance maps are close to orthogonal because, generally, no more than two end-members are mixed within one pixel. As opposed to the conventional norm-based sparse prior that assumes the abundance maps are independent, the proposed OSP explores the orthogonality between the abundance maps. Second, we propose the hyper-Laplacian loss to model the reconstruction error. The key observation is that the reconstruction error distribution usually has a heavy-tailed shape, which is better modeled by the hyper-Laplacian distribution rather than the commonly used Gaussian distribution. Third, to ease the side effect of outliers for end-member initializations, we develop a data-driven approach to detect outliers from the raw hyperspectral images. Extensive experiments on both synthetic and real-world data sets show that the proposed method significantly and consistently outperforms the compared state-of-the-art methods, with up to more than 50% improvements.**

***Index Terms*— Autoencoder, hyper-Laplacian distribution, outlier detection, sparse prior, spectral unmixing.**

I. Introduction

HYPERSPECTRAL images (HSIs) have been widely used in various remote sensing applications, including target detection, classification, and environmental monitoring [1]–[3]. However, due to the high spectral resolution of HSIs, their spatial resolutions are very limited compared with other optics operating systems in the visible spectrum domain. As a result, many pixels are mixed by several materials, causing difficulties in the characterization of hyperspectral data and understanding images [4]. For further hyperspectral applications, the end-members and their fractions should be extracted from data.

During the last decades, many algorithms have been proposed for the hyperspectral unmixing problem [1], [5]. Hyperspectral unmixing refers to the problem that given an HSI, a collection of pure constituent spectra, also named end-members, and a set of abundance fractions are extracted from the mixed pixel spectra [6]. The end-members are generally assumed to be pure materials, and the abundance fractions represent the percentage of each end-member.

Developing unmixing algorithms depends on specific mixing models, i.e., linear mixing models (LMMs) and nonlinear mixing models (NLMMs) [5]. LMMs assume that the mixed spectrum x is a linear combination of the end-member spectra e_m with additional reconstruction error η

$$x = \sum a_m e_m + \eta \tag{1}$$

where a_m is the abundance coefficient. NLMMs consider the multiple light reflections that occur at a multilayered or microscopic level. Recently, there has been a new trend for incorporating linear and nonlinear models for mixing [7]–[9]. Although NLMMs are more physically interpretable, many challenges still remain, including exploring more reliable metrics and prior knowledge. Therefore, most unmixing algorithms are LMM-based methods [10]. In this article, we build the unmixing algorithm based on LMMs.

Most unmixing methods can be classified into two categories: handcrafted-based methods [11]–[25] and learning-based methods [10], [26], [27]. Handcrafted methods, which derive estimations from the image itself without learning, are further classified into pure pixel-based methods and nonpure pixel-based methods, where the pure pixel refers to a single end-member. Pure pixel-based methods, such as vertex component analysis (VCA) [13] and N-FINDR [12], assume that pure pixels exist in the observed scene. These methods often use the orthogonality condition or projection on end-members in their estimation [11]. However, in real scenarios, pure pixels may not exist in the observed scene, particularly for those which have low spatial resolutions [14]. To overcome this problem, nonpure pixel-based methods are proposed, such as piecewise convex multiple-model (PCOMM) end-member detection [16], minimum volume-constrained nonnegative matrix factorization (MVCNMF) [15], robust collaborative nonnegative matrix factorization (RCNMF) [18], minimum volume simplex analysis [17], and others [19]–[25]. As the main trend in the

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last two decades, handcrafted methods have been shown to be effective in hyperspectral unmixing. However, these methods heavily depend on handcrafted designed priors, and they are not flexible compared with the learning-based methods.

Learning-based methods for hyperspectral unmixing have attracted much attention in recent years. Autoencoder, as an example of a learning-based method, has developed quickly in unmixing applications. Recently, due to their strong denoising and self-adaption capabilities, denoising autoencoders and nonnegative sparse autoencoders have been proposed to estimate the end-members and abundance fractions simultaneously. Ozkan *et al.* [26] proposed a shallow sparse autoencoder network, based on LMM, for end-member extraction and hyperspectral unmixing simultaneously. Palsson *et al.* [27] proposed a multilayer autoencoder with the spectral information divergence objective for the hyperspectral unmixing problem. Su *et al.* [10] combined the stacked autoencoder and variational autoencoder to conduct hyperspectral unmixing. All these works show the true potentials of autoencoders for hyperspectral unmixing tasks. In this article, we use the autoencoder to model the hyperspectral unmixing problem.

One major challenge in the hyperspectral unmixing problem is that both end-members and abundances are usually unknown, making the solution space too large to explore. Prior knowledge, therefore, should be introduced to restrict the solution space. Among various priors, sparse priors [28]–[31] for abundances are most popular. The sparse prior naturally follows the notion that only a few end-members are mixed within one pixel. For most of sparse-based works, the sparse prior is usually introduced using L_p norm($p \leq 1$) [28], [29]. However, as will be shown in this article, the conventional L_p norm prior assumes that the abundances are independent, i.e., it does not consider the relationships between abundances within one pixel. In this article, we propose a new prior that explores the orthogonality between abundances. The new prior, termed orthogonal sparse prior (OSP), is based on the key observation that the abundance maps of different end-members are close to orthogonal because there are often only one or two end-members in one pixel. Compared with the norm-based sparse prior, the OSP considers the relationships between the abundance maps, leading to a better result.

Another challenge is the presence of outliers in the HSIs. The presence of outliers would lead to end-member initialization problems, brining strong interference to the unmixing solutions [10]. However, many autoencoder-based methods only consider noise reduction and exhibit limitations when dealing with outliers. Su *et al.* [10] combined VCA and stacked autoencoders to generate good end-member initializations for further processing. This method first adopted VCA to obtain a set of end-member candidates, and these candidates were clustered into several training sets based on spectral angle distance (SAD). Finally, these training sets were used to train the stacked autoencoders to generate the end-member initializations. This method, however, is computationally expensive. In this article, we propose a simple data-driven approach to detect outliers from the raw HSI. The basic motivation is that outliers are usually different from their spatial neighborhoods. Specifically, we learn a data-guided map from the raw HSI, describing similarities between pixels and their neighborhoods. Given this data-guided map, outliers could be filtered out from the raw HSI using a simple strategy. The remaining pixels can be used to further conduct end-member initializations.

Many hyperspectral unmixing methods assume the reconstruction error, η, to be Gaussian white noise, which is equivalent to using the mean square error to measure differences between the ground truths and reconstructions [27], [32]. This is especially true for the nonnegative matrix factorization-based methods [8], [29], where they use the square of the Frobenius norm as the data fidelity term. Using the square of the Frobenius norm is equivalent to assuming that the reconstruction error of each matrix element follows the Gaussian white noise [33]. While the Gaussian distribution assumption is commonly used in the unmixing problem, as will be shown in this article, this assumption may not valid in real scenarios due to the presence of outliers and the low-rank structure of the reconstructions. In this article, we conduct an analysis of the distribution of reconstruction errors using several real-world HSI data sets with annotations. The results show that the reconstruction error distribution has significantly heavier tails than the Gaussian distribution. This observation indicates that the error distribution could be better modeled by the hyper-Laplacian distribution, which is also a heavy-tailed distribution, rather than the Gaussian distribution. Based on this fact, we propose the hyper-Laplacian loss to model the reconstruction error.

In summary, the main contributions of this article are threefold.

1) We propose an autoencoder with the OSP to simultaneously extract end-members and abundance maps from the raw HSIs.
2) We observe that the reconstruction error distribution could be better modeled by the hyper-Laplacian distribution rather than the commonly used Gaussian distribution. Based on our observations, we propose the hyper-Laplacian loss to model the reconstruction error.
3) We propose a simple but effective data-driven approach for outlier detection.

The rest of this article is organized as follows. In Section II, we introduce our method. In Section III, we use both synthetic and real-world HSI data sets to validate the effectiveness of the proposed method. Section IV concludes this article.

II. Proposed Method

A. Autoencoder With OSP

Let $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_n] \in R^{d \times n}$ be the matrix representation of a hyperspectral data set with d spectral bands and n pixels. Our goal is to extract $\mathbf{W}$ and $\mathbf{H}$ simultaneously from $\mathbf{X}$, where $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_m] \in R^{d \times m}$ is the end-member matrix containing m end-members, and $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_n] \in R^{m \times n}$ is the abundance matrix. We use an autoencoder to generate $\mathbf{H}$ and $\mathbf{W}$ from the raw HSI.

An autoencoder is a feedforward neural network that is trained to reproduce its input by learning the identity function. The autoencoder consists of two parts: an encoder and a

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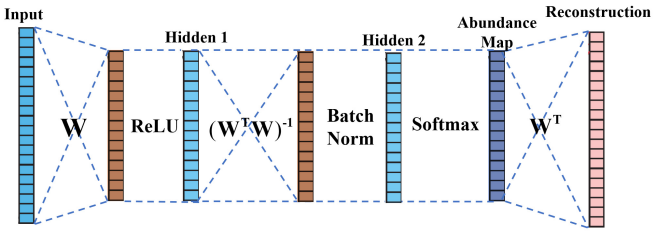


Fig. 1. Illustration of the proposed network structure.

decoder. The encoder encodes the input $\mathbf{x}_i$ to a hidden representation, which is the abundance vector $\mathbf{h}_i$ in this article. The decoder reconstructs the input vector using the hidden representation. In this article, the proposed autoencoder consists of a two-layer encoder and a one-layer decoder. The encoder has two fully connected layers. The first layer is followed by the ReLU activation function, while the second layer is followed by the batch normalization layer to alleviate optimization difficulties. On the top of the encoder, the Softmax operation is applied to obtain the abundances $\mathbf{H}$. Note that the Softmax operation naturally meets the nonnegative and sum-to-one constraints. The decoder is a fully connected layer without activation functions, where the parameters are the end-member matrix $\mathbf{W}^T$. Fig. 1 shows the proposed network structure.

As stated in Section I, because both the end-members and the abundances are unknown, priors should be introduced to restrict the solution space. Most of methods use norm-based sparse priors for abundance maps, assuming that there are only a few end-members mixed in one pixel. Take L_p $(p \leq 1)$ norm sparse prior as an example

$$L_1(\mathbf{h}_i) = \sum_j |h_{ij}|^p \tag{2}$$

where h_{ij} represents the jth element of $\mathbf{h}_i$. Minimizing (2) would lead to a sparse $\mathbf{h}_i$. However, this kind of prior assumes that abundances are independent. To understand this point, we derive this prior using the probability framework. By assuming that $h_{ij}(j = 1, \ldots, m)$ are independent, the probability density $P(\mathbf{h}_i)$ is

$$P(\mathbf{h}_i) = \prod_j P(h_{ij}) \tag{3}$$

where $P(h_{ij})$ conforms to the Laplace or hyper-Laplace distribution, i.e., $P(h_{ij}) \propto e^{-|h_{ij}|^p} (p \leq 1)$. By taking log and negative-sign operations, we can easily derive that maximizing (3) is equivalent to minimizing (2), and vice versa. Thus, the underlying assumption of (2) is the independences between abundances.

We can also derive the underlying independence assumption from an optimization perspective. The derivative, with respect to h_{ij}, is

$$\frac{\partial L_1}{\partial h_{ij}} = \frac{\text{sgn}(h_{ij})}{h_{ij}^{1-p}} \tag{4}$$

where sgn is the sign function. Equation (4) shows that the iteration of h_{ij} only relies on itself, i.e., it does not consider interactions between abundance maps.

In this article, we propose a new sparse prior, called OSP, to explore the interactions between the abundance maps. The basic motivation is that because the abundance vector is very sparse, it often contains no more than two nonzero elements, i.e., the top two main end-members usually contribute more than 90% proportions in one pixel, and the rest end-members occupy only a tiny portion which can be ignored. Therefore, the arbitrary two different abundance maps are close to orthogonal.

We use three well-annotated data sets, i.e., Urban, Jasper Ridge, and Samson data sets, to verify the reasonability of the proposed assumption. Fig. 2 shows the abundance histograms over three data sets, where the horizontal axis represents the summation of the top two abundance coefficients. The larger the summation, the larger portion the top two end-members contribute. We can see that the pixels, in which the top two main end-members contribute over 90% abundance, make up the majority of data sets (80.6%, 82.5%, and 92% in Urban, Jasper Ridge, and Samson data sets, respectively). This statistic gives strong support to the proposed assumption. Fig. 3 further shows an example, where there are five ground-truth abundance maps as well as the visualization of their correlation coefficient matrix $\mathbf{M}^{\text{cor}}$, whose element is computed as follows:

$$\mathbf{M}_{ij}^{\text{cor}} = \frac{\mathbf{H}_{i\cdot} \cdot \mathbf{H}_{j\cdot}}{\|\mathbf{H}_{i\cdot}\|_2 \|\mathbf{H}_{j\cdot}\|_2} \tag{5}$$

where $\mathbf{H}_{i\cdot}$ represents the ith row of $\mathbf{H}$. The bright pixels in one abundance map are very likely to be dim in the remaining abundance maps; thus, they are almost orthogonal to each other. The correlation matrix is also close to diagonal, which further supports our analysis.

Mathematically, given a batch of generated abundance maps $\mathbf{B} \in R^{b \times m}$, where b is the batch size and m is the number of end-members, the OSP is

$$L_{\text{OSP}}(\mathbf{B}) = \sum_{i<j} \frac{\mathbf{B}_{\cdot i} \cdot \mathbf{B}_{\cdot j}}{\|\mathbf{B}_{\cdot i}\|_2 \|\mathbf{B}_{\cdot j}\|_2} \tag{6}$$

where $\mathbf{B}_{\cdot i}$ represents the ith column of $\mathbf{B}$. Note that the minimum value of $\mathbf{B}_{ij}$ is zero; thus, the minimizer of L_{OSP} is zero as well. As will be shown in the ablation study, the OSP leads to the sparser abundance maps than the conventional norm-based sparse prior.

B. Hyper-Laplacian Loss

We optimize the proposed autoencoder by minimizing the following objective function E:

$$\begin{aligned} E = f(\mathbf{X}, \mathbf{R}) &- \sum_i \log\left(\frac{\mathbf{R}_{\cdot i} \cdot \mathbf{X}_{\cdot i}}{\|\mathbf{R}_{\cdot i}\|_2 \|\mathbf{X}_{\cdot i}\|_2}\right) + \rho_1 \ L_{\text{OSP}} \\ &+ \rho_2 \sum_i L_1(\mathbf{h}_i) - \rho_3 \ S(\mathbf{M}_{\mathbf{W}<0} \odot \mathbf{W}) \\ &+ \rho_4 \ S(\mathbf{M}_{\mathbf{W}>1} \odot \mathbf{W}) \end{aligned} \tag{7}$$

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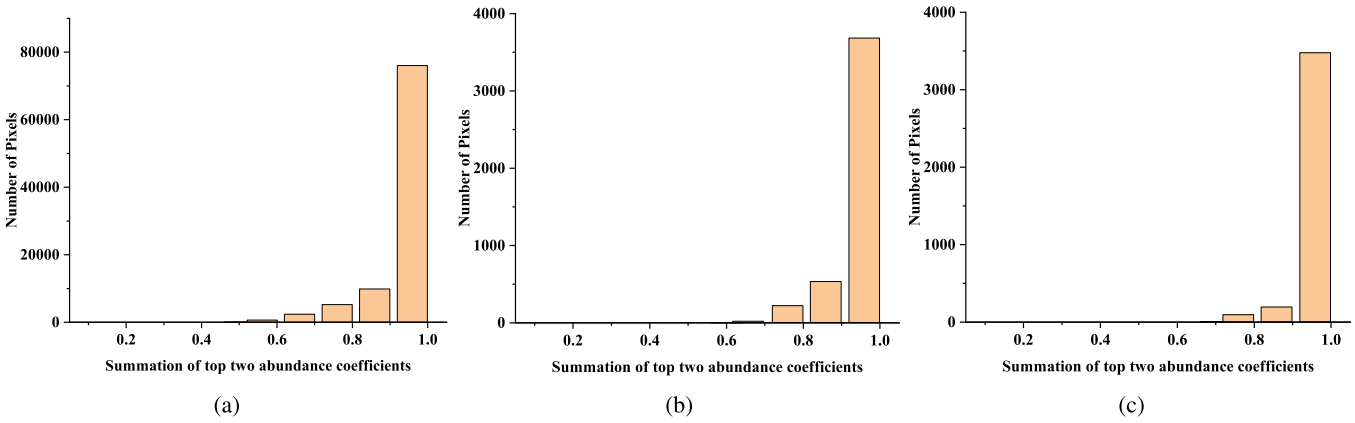


Fig. 2. Histograms of summations of top two abundance coefficients. Each bin represents 0.1 abundance. (a) Urban data set. (b) Jasper Ridge data set. (c) Samson data set.

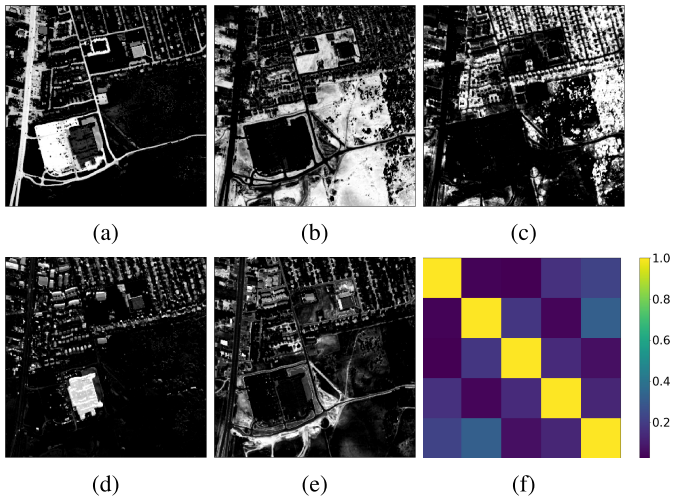


Fig. 3. Ground-truth abundance maps and their correlation coefficient matrix. (a)–(e) Five ground-truth abundance maps. (f) Visualization of the correlation coefficient matrix—the brighter value indicates the stronger correlation between two maps.

where $\mathbf{X}_{\cdot i}$ represents the ith column of $\mathbf{X}$, $\mathbf{R} = \mathbf{WH}$, and

$$(\mathbf{M}_{\mathbf{W}<0})_{ij} = \begin{cases} 1, & \text{if } W_{ij} < 0 \\ 0, & \text{else} \end{cases} \tag{8}$$

$$(\mathbf{M}_{\mathbf{W}>1})_{ij} = \begin{cases} 1, & \text{if } W_{ij} > 1 \\ 0, & \text{else} \end{cases} \tag{9}$$

$$S(\mathbf{M}) = \sum_{ij} M_{ij}. \tag{10}$$

$\odot$ is the elementwise multiplication. ρ_1, ρ_2, ρ_3, and ρ_4 are the balancing weights. The first term on the right-hand side of (7), f, is the data-fidelity term, the second term controls angle similarities between the reconstructions and the inputs, and the last two terms restrict W_{ij} into $[0, 1]$.

The exact form of f relies on the distribution of the reconstruction error η. Most methods assume that η follows the Gaussian distribution, indicating that f is the mean square error. However, as we will demonstrate, due to the heavy-tailed shape of the error distribution, it is not appropriate to model η as the Gaussian distribution. Fig. 4 shows an HSI and histogram of the reconstruction error. We see that the distribution has most of its mass on small values but gives significantly more probability to large values than the Gaussian distribution.

To understand this observation, we explain what factors contribute to the heavy tails. First, the heavy tails can come from the outliers in the HSI. In a real situation, the HSI is likely to contain outliers due to various reasons, such as shading, atmospheric effect, material aging, and additional light sources. Second, the rank of the reconstruction, rank($\mathbf{WH}$), is much lower than the raw HSI. For the reconstruction, we have rank($\mathbf{WH}$) $\leq$ max(rank($\mathbf{W}$), rank($\mathbf{H}$)) $\leq m$, where m is the number of end-members, while for the raw HSI, the rank typically ranges from 10 to 50, according to the complexity of the scene (we collected 20 HSIs and computed their ranks, and the rank was computed by ignoring the eigenvalues whose values are below 1). The low-rank structure makes reconstructions have very different appearances compared with the raw HSIs. Fig. 5 shows an example in which we select some representative bands. We see that the reconstruction has many more distinguishable visual features than the raw HSI, resulting in the heavy tails of the error distribution.

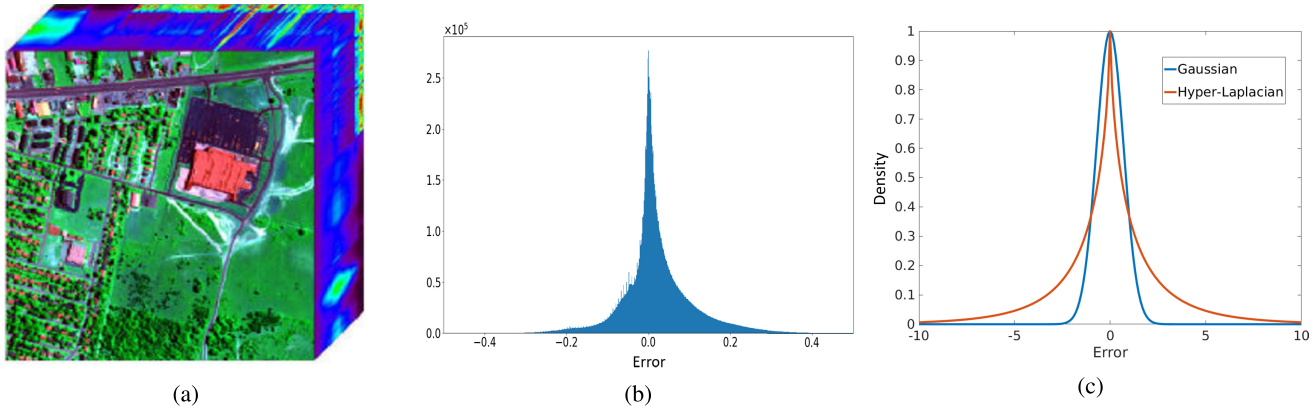
In this article, we use the hyper-Laplacian distribution, which also has heavy tails, to model the error distribution. Fig. 4(c) shows the plots of the density function of the hyper-Laplacian distribution and the Gaussian distribution. Formally, the hyper-Laplacian distribution has the following form:

$$P(\mathbf{x}) \propto e^{-k|\mathbf{x}|^p} \tag{11}$$

where $p \in [0.5, 1)$ and k is a constant. Maximizing (11) is equivalent to minimizing $\|\mathbf{x}\|_p^p$. Thus, the fidelity term f is

$$f(\mathbf{X}, \mathbf{WH}) = \|\mathbf{X} - \mathbf{WH}\|_p^p. \tag{12}$$

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(a) (b) (c)

Fig. 4. One example of the reconstruction analysis. (a) Test HSI. (b) Histogram of the reconstruction error. (c) Probability density of the Gaussian and the hyper-Laplacian distributions.



Fig. 5. Visual comparisons of the reconstructions and the raw HSI. First row: four reconstructed bands using the ground-truth end-members and abundance maps. Second row: corresponding bands of the raw HSI.

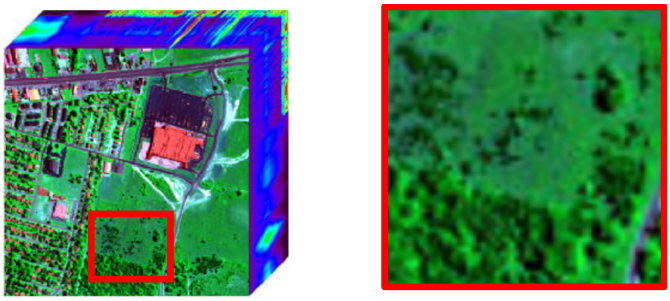


Fig. 6. Zoomed-in example for the transition area. The transition area is more likely to contain highly mixed pixels.

We call (12) as the hyper-Laplacian loss. Hence, the exact form of E is

$$E = \|\mathbf{X}-\mathbf{R}\|_p^p - \sum_i \log\left(\frac{\mathbf{R}_{\cdot i}\cdot\mathbf{X}_{\cdot i}}{\|(\mathbf{WH})_{\cdot i}\|_2\|\mathbf{X}_{\cdot i}\|_2}\right) + \rho_1\ L_{\text{OSP}} + \rho_2\sum_i L_1(\mathbf{h}_i) - \rho_3\ S(\mathbf{M}_{\mathbf{W}<0}\odot\mathbf{W}) + \rho_4\ S(\mathbf{M}_{\mathbf{W}>1}\odot\mathbf{W}). \tag{13}$$

C. Outlier Detection and Parameter Initialization

Because E is nonconvex, the optimization process could benefit from a good initialization of $\mathbf{W}$. However, outliers in the HSI would lead to a poor initialization and strongly interfere with the unmixing solutions. Thus, it is necessary to remove outliers before end-member initializations. Note that we treat highly mixed pixels as outliers as well.

We observe that outliers usually have different appearances compared with their neighborhood pixels. This is also true for the highly mixed pixels since most of them lie in the transition areas. Fig. 6 illustrates an example, where there are two end-members in the scene. The pixels in the transition area are likely to be highly mixed by their neighboring pixels; therefore, they are treated as outliers. Based on this fact, a data-guided map can be learned by measuring the uniformity of the neighboring pixels over the entire image, and the nonuniform areas are likely to contain outliers. Specifically, for the ith pixel, its value in the data-guided map d_i could be estimated as follows:

$$d_i = \frac{1}{|\mathcal{N}_i|}\sum_{j\in\mathcal{N}_i} s_{ij} \tag{14}$$

where $\mathcal{N}_i$ is the neighborhood of the ith pixel that includes $|\mathcal{N}_i|$ surrounding pixels, and s_{ij} is the similarity measurement between the ith pixel and the jth pixel. In this article, we use the heat kernel similarity metric

$$s_{ij} = e^{-\frac{\|\mathbf{x}_i-\mathbf{x}_j\|_2^2}{\sigma}} \tag{15}$$

where $\sigma = 0.005$ controls the contrast of the data-guided map. Fig. 7 shows an example of a data-guided map and the detected outliers, where the dim value represents the large probability of outliers. We see that most detected outliers are either in the transition areas or the isolated points, which conforms to our assumption. With a learned map, we can set a threshold τ to separate the pixels into outliers and normal pixels. The pixels whose values are smaller than τ are considered to be outliers, while those whose values are larger than τ are considered to be normal pixels. Finally, the normal pixels are used to conduct the end-member initializations. In this article, we set τ to 0.5 and use Distance-MaxD (DMaxD) [34] to initialize the end-member matrix $\mathbf{W}$.

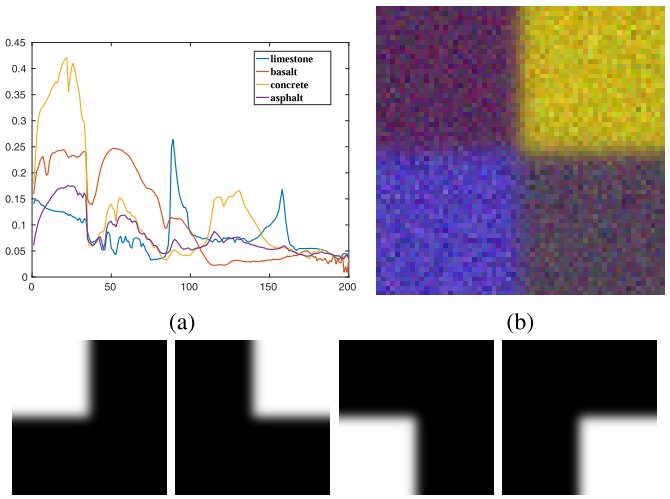
After initializing $\mathbf{W}$, we further propose a new initialization method for the encoder. The proposed method is motivated by

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(a) (b) (c)

Fig. 7. Data-guided map and detected outliers of the Urban hyperspectral data set. (a) Test data set. (b) Data-guided map. (c) Detected outliers.



(a) (b)

(c)

Fig. 8. Synthetic data set. (a) Original spectra from the ASTER library. (b) Pseudocolor image. (c) Ground-truth abundance maps.

the analytic solution of least square methods. Given the endmember matrix $\mathbf{W}$, the solution of $\|\mathbf{X}-\mathbf{WH}\|_2^2$ is

$$\mathbf{H}=(\mathbf{W}^T\mathbf{W})^{-1}\mathbf{W}^T\mathbf{X}. \tag{16}$$

Equation (16) can be viewed as the stack of two fully connected layers, where $\mathbf{W}$ and $(\mathbf{W}^T\mathbf{W})^{-1}$ represent the parameter matrix of the first layer and the second layer, respectively. We use these two matrices as the initialization of our encoder.

III. Experiments

In this section, we evaluate the performance of the proposed method on several data sets. We use both synthetic and real-world data sets for testing. First, we give the implementation details of the proposed method, and then, we summarize the details of the HSI data sets, baseline methods, and evaluation metrics. In order to perform reliable assessments, we repeat the experiments ten times and report the mean and standard deviation of the results. We also give visual comparisons of abundance maps. Finally, we conduct an ablation study to evaluate each component of the proposed method, analyze hyperparameter sensitivity, and compare computing time with the other state-of-the-art methods.

A. Implementation Details

We implement the proposed method in Pytorch. For all the experiments, we use Adam optimizer with a learning rate 0.001 and momentum 0.9. The batch size is 256. We set

TABLE I

SAD ($\times 10^{-2}$) and RMSE ($\times 10^{-2}$) From the Synthetic Data Set. Mean and Standard Deviation Are Reported. Best Results Are Reported in Bold

Endmember(SAD)	VCA	PCOMM	MLM	DMaxD	AEU	Proposed
#1	6.96±0.7	1.14±0	6.75±0	14.67±0	1.25±0.2	**0.78±0.1**
#2	4.04±0.1	0.27±0	5.39±0	8.48±0	0.86±0.1	**0.79±0.3**
#3	5.5±0.7	1.49±0	10.36±0	6.24±0	0.97±0.1	**0.69±0.1**
#4	3.33±0.3	3.16±0	26.33±0	28.8±0	4.92±0.8	**1.3±0.1**
Average	4.96±0.2	1.52±0	12.21±0	14.6±0	2±0.2	**0.89±0.1**
Abundance(RMSE)	VCA	PCOMM	MLM	DMaxD	AEU	Proposed
#1	**4.61±1.2**	27.76±0	18.76±0	24.4±0	7.81±2.2	4.91±0.9
#2	**3.17±0.5**	19.82±0	10.31±0	10.88±0	8.44±1.1	4.87±0.9
#3	6.53±0.5	25.24±0	18.54±0	23.08±0	5.18±0.9	**4.04±0.8**
#4	17.6±2.2	38.78±0	31.67±0	39.06±0	17.11±3.3	**6.75±0.4**
Average	7.99±0.4	27.9±0	19.82±0	24.36±0	9.63±1.7	**5.14±0.6**

TABLE II

SAD ($\times 10^{-2}$) and RMSE ($\times 10^{-2}$) From the Urban Data Set. Mean and Standard Deviation Are Reported. Best Results Are Reported in Bold

Endmember(SAD)	VCA	PCOMM	MLM	DMaxD	AEU	Proposed
#1	21.89±3.2	11.35±0	12.96±0	14.8±0	11.96±4.1	**8.11±1.1**
#2	38.91±4.7	29.45±0	33.31±0	24.8±0	25.14±7.7	**8.68±1.1**
#3	26.77±8.1	10.32±0	11.11±0	**5.38±0**	10.02±2.4	6.15±0.5
#4	77.69±3.3	22.45±0	15.03±0	17.4±0	9.20±4.2	**6.01±1.3**
#5	82.69±2.3	8.12±0	8.96±0	16.6±0	7.34±1.8	**5.17±0.9**
Average	49.59±3.9	16.34±0	16.28±0	15.8±0	12.93±4.5	**6.83±1.7**
Abundance(RMSE)	VCA	PCOMM	MLM	DMaxD	AEU	Proposed
#1	31.34±5.3	24.65±0	32.09±0	31.7±0	23.96±6.4	**16.55±2.1**
#2	39.91±5.2	29.52±0	34.64±0	31.1±0	23.20±3.6	**15.91±1.6**
#3	28.89±4.2	26.68±0	27.33±0	36.1±0	18.25±2.5	**15.59±2.3**
#4	21.42±2.5	25.27±0	17.24±0	21.5±0	16.59±4.6	**10.43±1.1**
#5	31.52±4.2	16.12±0	18.81±0	18.9±0	14.12±4.9	**14.11±1.1**
Average	30.62±3.9	24.45±0	26.02±0	27.9±0	19.22±4.7	**14.52±0.6**

$|\mathcal{N}|=8$, $p=0.7$, $\rho_1=0.5$, $\rho_2=1$, and $\rho_3=\rho_4=10$ throughout this article. We train the encoder and decoder alternatively; at the beginning of the training, we fix the parameters in the decoder and train the encoder for 2 epochs, and then, we fix the encoder and train the decoder for 1 epochs. This process is repeated until the number of epochs reaches 40.

B. Baselines and Evaluation Metrics

1) Baselines: We compare the proposed method with five hyperspectral unmixing algorithms.

1) *VCA [13]:* The code is available at http://www.lx.it.pt/ bioucas/code/demo_vca.zip.
2) *PCOMM Unmixing [16]:* The code is available at https://github.com/GatorSense/PCOMM.
3) *Multilinear Unmixing Model (MLM) [7]:* The code is available at https://github.com/qw245/MLMp.
4) *DMaxD [34]:* The code is available at https://sites.google.com/site/robheylenresearch/code/DMaxD.tgz?attredirects=0&d=1.
5) *Autoencoder-Based Unmixing (AEU) [27]:* Because the code is not available from the authors, we implement it ourselves.

2) Metrics: We use SAD and root-mean-square error (RMSE) to measure the unmixing performances

$$\mathrm{SAD}(\mathbf{w}_i,\mathbf{w}_j)=\arccos\left(\frac{\mathbf{w}_i\cdot\mathbf{w}_j}{\|\mathbf{w}_i\|_2\|\mathbf{w}_j\|_2}\right) \tag{17}$$

$$\mathrm{RMSE}(\mathbf{x},\mathbf{y})=\sqrt{\frac{1}{N}\|\mathbf{x}-\mathbf{y}\|_2^2}. \tag{18}$$

Smaller values indicate better performances for both metrics.

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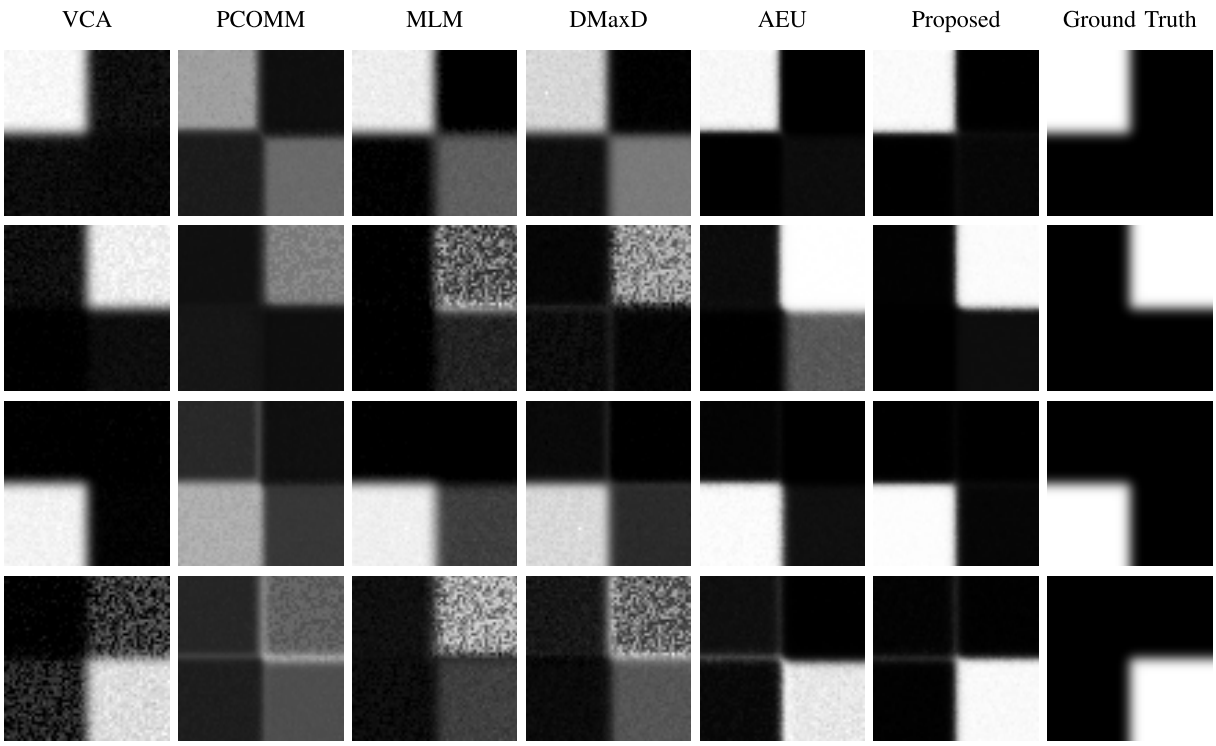


Fig. 9. Visual comparisons of the abundance maps of different methods and the ground truths from the synthetic data set. From the first column to the last column: VCA, PCOMM, MLM, DMaxD, AEU, the proposed method, and the ground truths. From the first row to the last row: limestone, basalt, concrete, and asphalt.



Fig. 10. Pseudocolor images of the test data sets. (a) Urban. (b) Jasper Ridge. (c) Samson. (d) Washington DC.

TABLE III

SAD ($\times 10^{-2}$) AND RMSE ($\times 10^{-2}$) FROM THE JASPER RIDGE DATA SET. MEAN AND STANDARD DEVIATION ARE REPORTED. BEST RESULTS ARE REPORTED IN BOLD

Endmember(SAD)	VCA	PCOMM	MLM	DMaxD	AEU	Proposed
#1	68.87±5.1	12.37±0	9.16±0	3.3±0	8.41±3.2	**3.78±0.8**
#2	22.7±0.9	6.1±0	19.41±0	25.43±0	9.2±3.1	**4.63±0.3**
#3	24.08±5.9	24.7±0	11.44±0	16.32±0	12.02±5.4	**7.85±3.7**
#4	23.09±3.4	21.82±0	17.68±0	5.64±0	18.20±6.7	**2.88±0.5**
Average	34.69±3.2	16.0±0	14.42±0	12.7±0	11.95±5	**4.78±1.9**
Abundance(RMSE)	VCA	PCOMM	MLM	DMaxD	AEU	Proposed
#1	34.39±3.45	23.45±0	15.70±0	19.33±0	13.61±3.4	**5.86±1.4**
#2	19.4±2.2	**7.84±0**	15.71±0	14.39±0	8.52±2.3	7.91±1.8
#3	13.75±1.2	21.02±0	11.75±0	21.27±0	13.56±3.7	**7.58±0.6**
#4	21.6±3.1	14.97±0	8.77±0	22.37±0	22.31±5.6	**7.15±0.8**
Average	22.28±2.3	16.82±0	12.98±0	15.5±0	14.5±4.2	**7.12±1.1**

C. Synthetic Data Set

The synthetic image is of size 60 × 60 pixels. There are four end-members: limestone, basalt, concrete, and asphalt. Following [35], we simulate the real scenario by increasing end-member variabilities. Specifically, the end-members are sampled from the Gaussian mixture models (GMMs), the means of which are sampled from the ASTER spectral library [36]. The four materials occupy the four quadrants of the square image. We use the Gaussian smoothing to make the boundary pixels of each quadrant be mixed by their neighboring materials and add the Gaussian noise with 0.001 variance to the image. The spectral signatures, the pseudocolor image, and the ground-truth abundance maps are shown in Fig. 8.

Table I shows the performances of the six methods, where there are two subtables that show SADs and RMSEs for different methods. Each row of the subtable represents the performance of one end-member, except for the last row that shows the average performance. We see that the proposed method achieves the best performances for all metrics, outperforming the other state-of-the-art methods by a large margin. Specifically, the proposed method reduces the SAD error by almost a factor of two compared with the second-best method PCOMM. For the abundance comparisons, the proposed method produces the best results for #3 and #4 and achieves the comparable performance for #1 and #2. In terms of the average RMSE, the proposed method reduces 38.7% for the average RMSE compared with the second-best method.

We further visualize the estimated abundance maps using different methods and the ground truths in Fig. 9. We can see that all the compared methods have some difficulties with some specific end-members, e.g., VCA, PCOMM, MLM, and DMaxD fail to estimate the pure pixels of asphalt, and AEU cannot fully separate basalt and asphalt. In contrast, the proposed method correctly extracts different end-members from the raw HSI.

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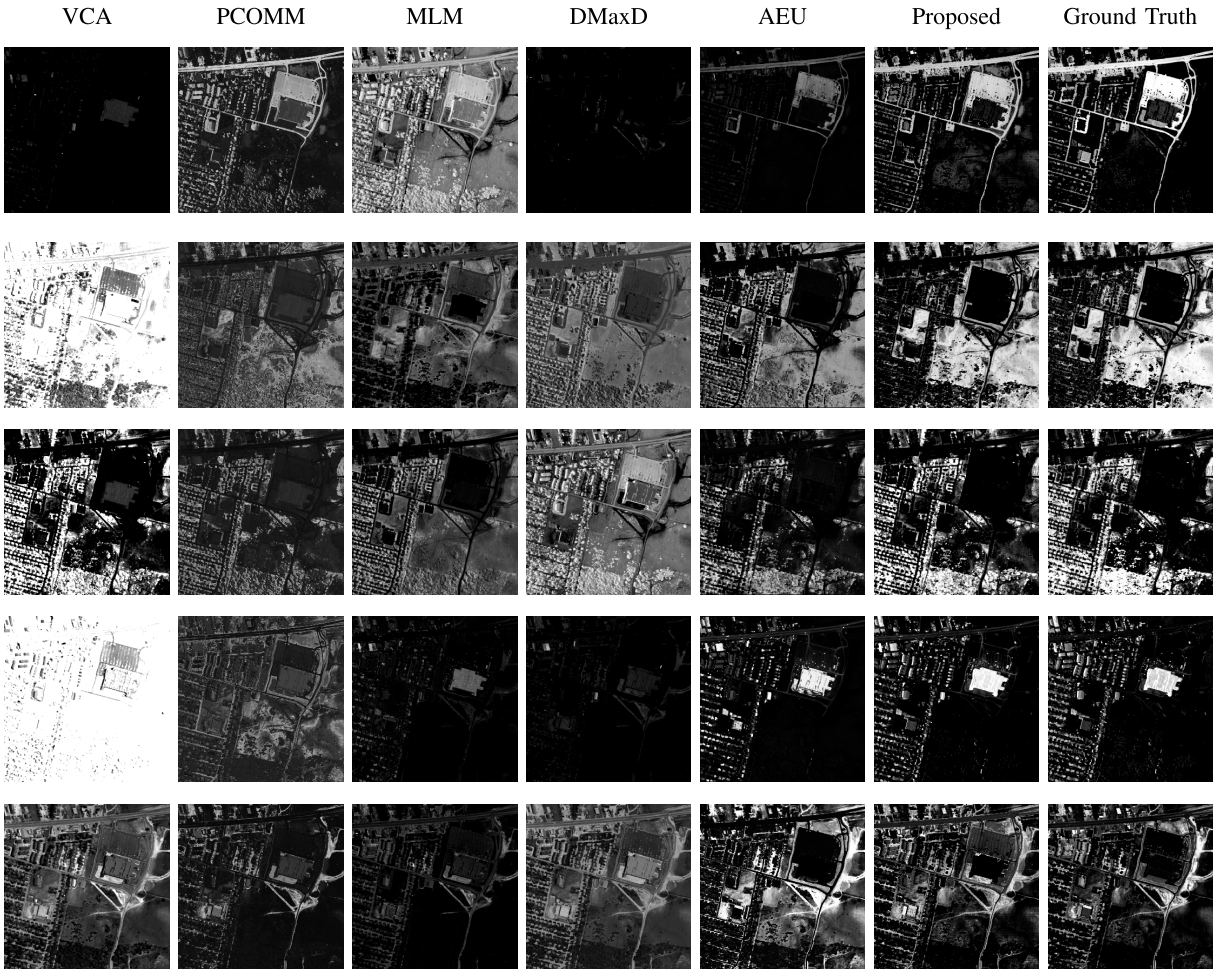


Fig. 11. Visual comparisons of the abundance maps of different methods and the ground truths of the Urban data set. From the first column to the last column: PCOMM, MLM, AEU, the proposed method, and the ground truths. From the first row to the last row: asphalt, grass, tree, roof, and dirt.

D. Real-World Data Sets

We use four real-world data sets to evaluate the proposed method, which are listed in the following.

1) *Urban:* The data have 307 × 307 pixels. Due to the water-vapor absorption and atmospheric effects, we remove 1–4, 76, 87, 101–111, 136–153, and 198–210 channels. There are five constituent materials: Asphalt (#1), Grass (#2), Tree (#3), Roof (#4), and Dirt (#5).
2) *Samson:* Data are generated by the SAMSON sensor. It contains 156 channels and the spectral ranges from 0.4 to 0.9 μm. Following previous work [31], [32], [37], a subimage of the original data (95 × 95) is considered. There are three material types: Rock (#1), Tree (#2), and Water (#3).
3) *Jasper Ridge:* Data are captured by the AVIRIS sensor. A subimage of 100 × 100 pixels is used. Due to the atmospheric effects and water-vapor absorption, we remove 1–3, 108–112, 154–166, and 220–224 channels. Four materials are observed in the scene: Tree (#1), Water (#2), Dirt (#3), and Road (#4).
4) *Washington DC:* Data are captured from the Washington DC Mall. It has 191 bands covering the wavelength ranging from 0.4 to 2.4 μm. Due to the large size of the raw HSI, we crop a subimage with a size of 256 × 256. Five materials are observed in the scene: soil (#1), grass (#2), water & shadow (#3), asphalt (#4), and tree (#5).

Fig. 10 shows the pseudocolor images of these data sets.

E. Model Comparisons

Because the ground-truth end-members and abundances are valid for the Samson, Jasper Ridge, and Urban data sets, we show both quantitative and qualitative comparisons for these data sets. For Washington DC data set, since the ground truths are invalid, we only show the qualitative comparisons.

The quantitative results are summarized in Tables II–IV. We see that the proposed method generally outperforms the other state-of-the-art methods by a large margin, and in a few cases, it achieves comparable results with the best results of other methods.

Table II shows the performances of six methods on Urban. In terms of the performances for single end-member, our method outperforms the other methods by a large margin, reducing up to 65% SAD and 31.4% RMSE, respectively, compared with the second-best method. In terms of the average performance comparison, the proposed method is

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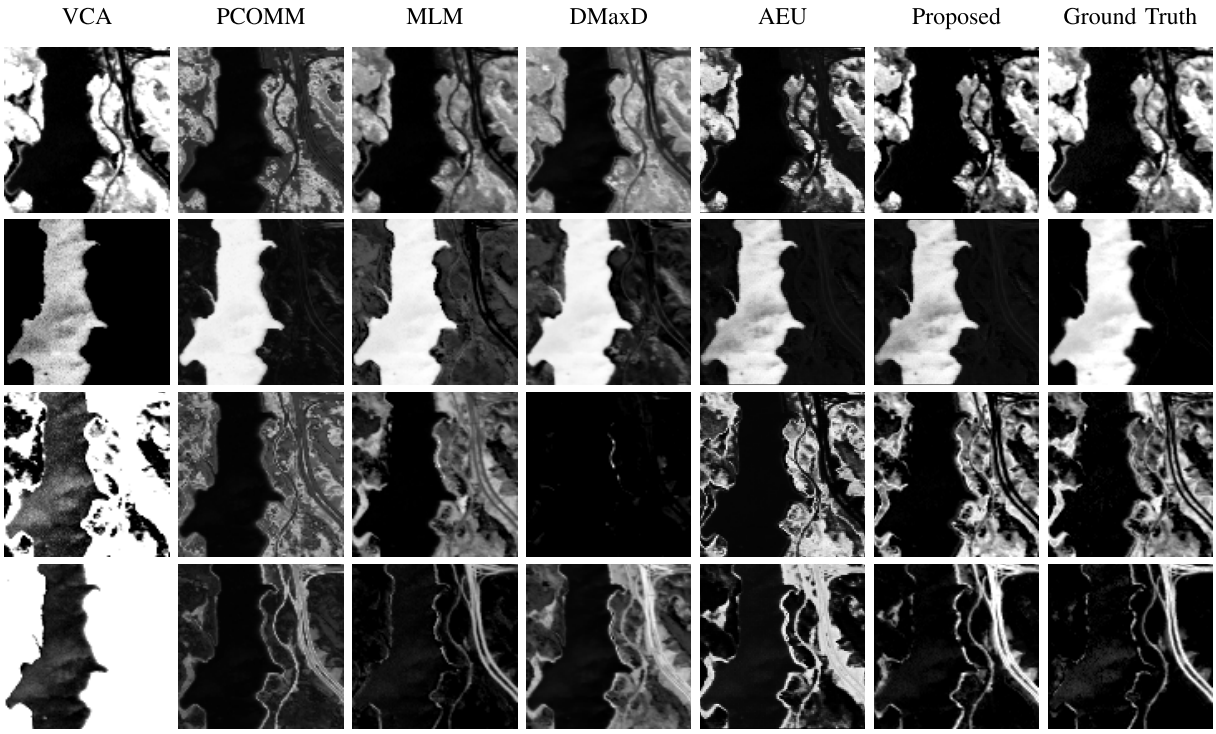


Fig. 12. Visual comparisons of the abundance maps of different methods and the ground truths from the Jasper Ridge data set. From the first column to the last column: PCOMM, MLM, AEU, the proposed method, and the ground truths. From the first row to the last row: tree, water, dirt, and road.

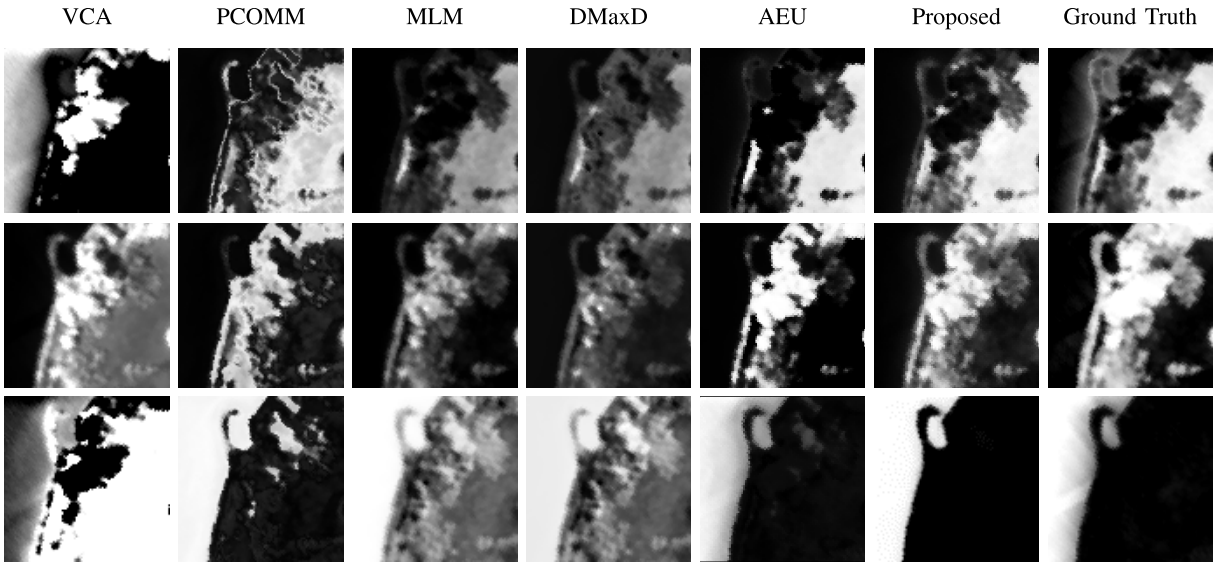


Fig. 13. Visual comparisons of the abundance maps of different methods and the ground truths from the Samson data set. From the first column to the last column: PCOMM, MLM, AEU, the proposed method, and the ground truths. From the first row to the last row: rock, tree, and water.

the best method that reduces 47.2% for SAD and 24.5% for RMSE according to the results of the second-best method.

The results on the Jasper Ridge are illustrated in Table III. The proposed method produces the best results for the end-member comparisons, reducing the SAD error by a factor of two. For the abundance comparisons, the proposed method produces the best results for #1, #3, and #4 and achieves comparable performances for #2. The proposed method also reduces 50.9% for the average RMSE compared with the second-best method.

The results on the Samson are summarized in Table IV. We see that apart from the proposed method, AEU shows better results than the others, indicating the good potential of the autoencoder for the hyperspectral unmixing task. However, our model obtains the best performance, reducing 11.1% and 12.1% for the average SAD and RMSE, respectively, compared with AEU.

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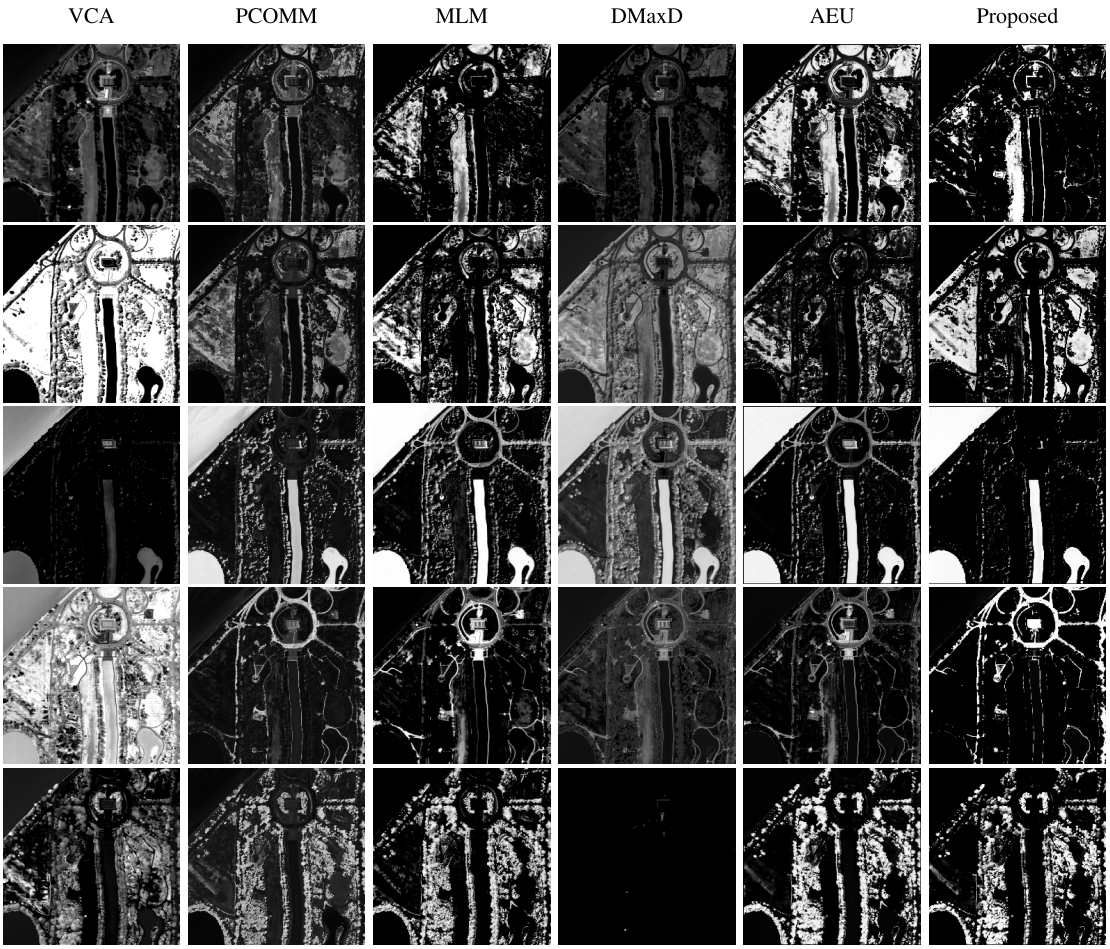


Fig. 14. Visual comparisons of the abundance maps of different methods from the Washington DC data set. From the first column to the last column: VCA, PCOMM, MLM, AEU, and the proposed method. From the first row to the last row: soil, grass, water & shadow, asphalt, and tree.

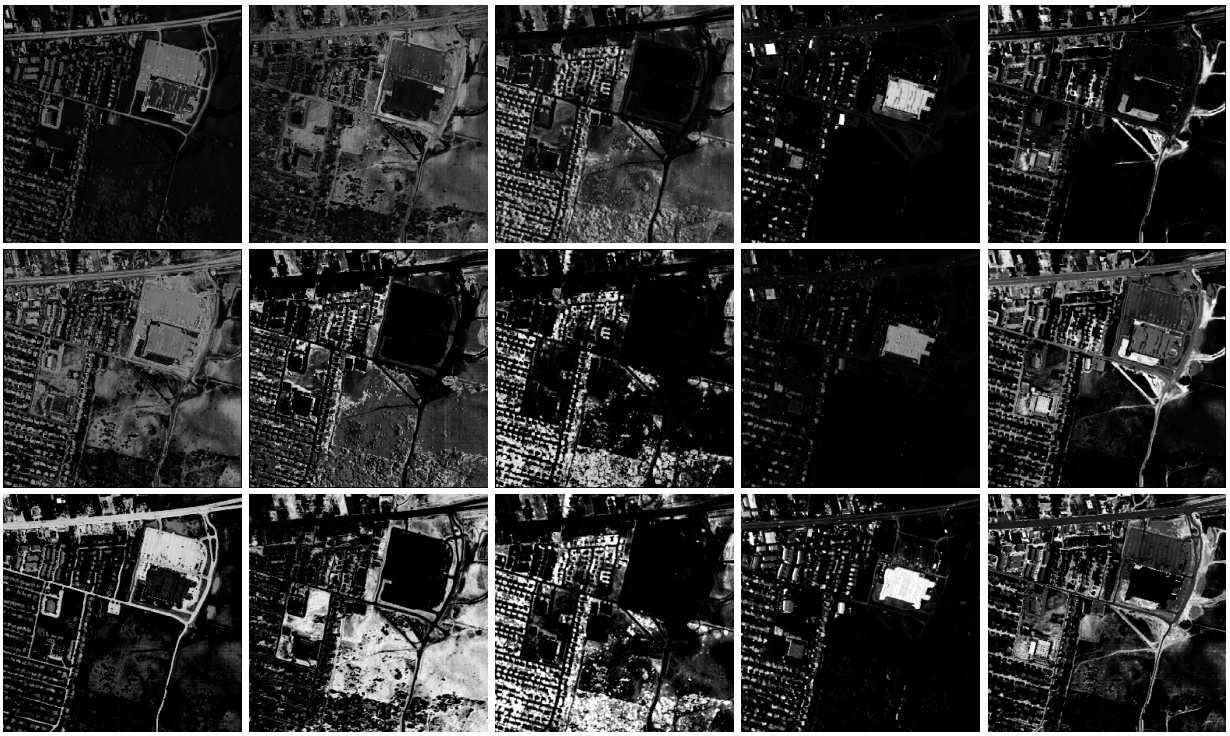
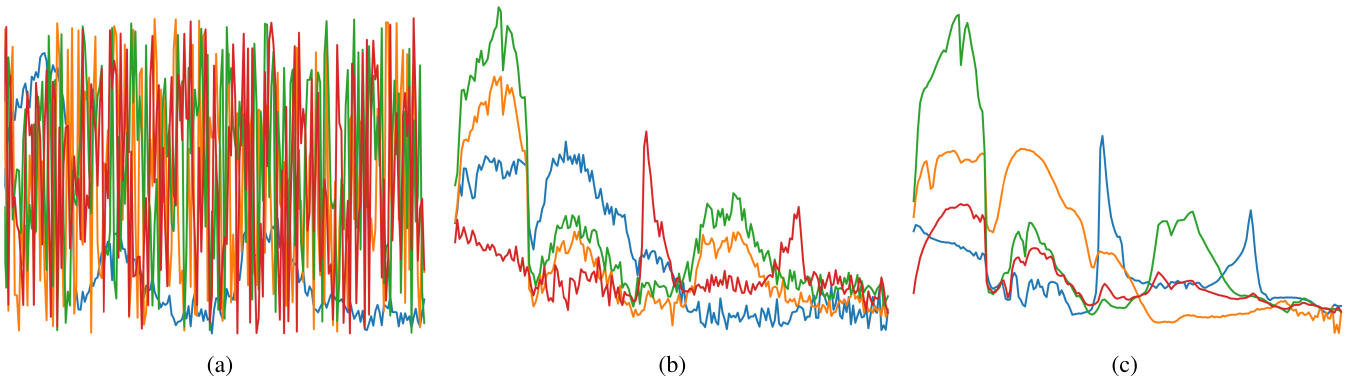


Fig. 15. Effectiveness of OSP and hyper-Laplacian loss. First row: results obtained without OSP. Second row: results obtained with MSE. Third row: results obtained with OSP and hyper-Laplacian loss.

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(a) (b) (c)

Fig. 16. End-member initializations with and without the outlier detection process. (a) Initialized end-members without the outlier detection process. (b) Initialized end-members with the outlier detection process. (c) Ground-truth end-members.

TABLE IV

SAD ($\times 10^{-2}$) AND RMSE ($\times 10^{-2}$) FROM THE SAMSON DATA SET. MEAN AND STANDARD DEVIATION ARE REPORTED. BEST RESULTS ARE REPORTED IN BOLD

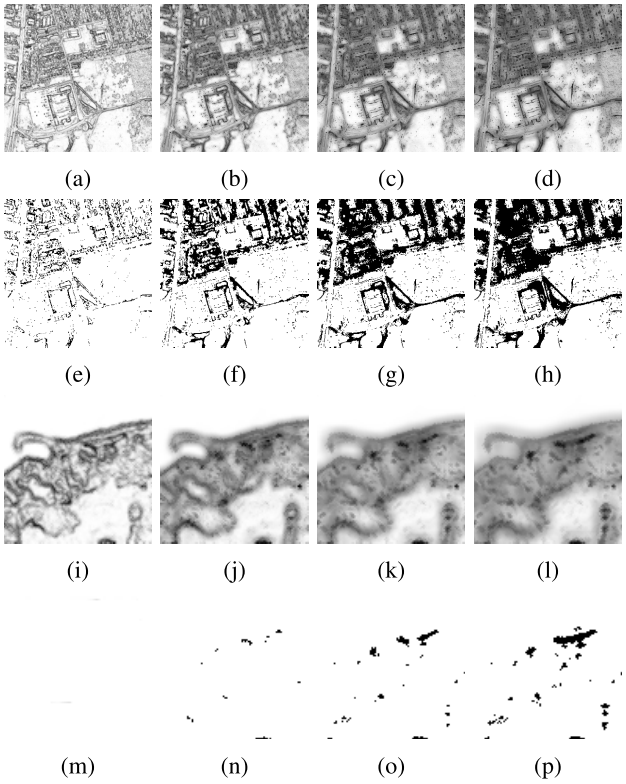
Endmember(SAD)	VCA	PCOMM	MLM	DMaxD	AEU	Proposed
#1	22.33±3.1	3.13±0	3.27±0	4.04±0	2.78±1.5	**0.72±0.01**
#2	4.96±0.7	4.97±0	5.65±0	5.95±0	4.44±0.6	**4.37±0.4**
#3	12.87±0.4	16.87±0	21.1±0	78.79±0	6.37±1.2	**5.22±1.9**
Average	13.39±1.1	14.0±0	10.0±0	29.36±0	3.86±1.6	**3.43±0.8**
Abundance(RMSE)	VCA	PCOMM	MLM	DMaxD	AEU	Proposed
#1	21.92±6.3	19.69±0	23.32±0	25.42±0	16.1±1.9	**13.19±1.2**
#2	29.64±3.3	24.74±0	28.28±0	32.96±0	13.43±1.7	**12.27±1.3**
#3	32.42±0.6	24.17±0	42.95±0	40.43±0	6.79±0.8	**6.48±0.7**
Average	19.66±3.2	22.86±0	31.52±0	19.76±0	12.11±6.2	**10.64±1.1**

TABLE V

EFFECTS OF EACH COMPONENT IN OUR DESIGN. THE VALUES IN THE TABLE ARE MEAN SAD ($\times 10^{-2}$)

OSP	Hyper-Laplacian Loss	Outlier Detection	#1	#2	#3	#4	#5	Average
			17.3	39.7	15.9	28.9	7.09	21.8
√			8.55	13.9	11.9	6.26	5.24	9.18
	√		20.8	14.2	19.5	6.72	14.4	15.1
		√	10.5	32.1	17.5	26.6	4.66	18.3
√	√		8.43	9.22	8.54	5.08	6.22	7.49
√		√	8.81	8.75	8.81	7.62	5.23	7.84
	√	√	13.3	10.7	17.1	3.18	14.2	11.7
√	√	√	**8.11**	**8.68**	**6.15**	**6.01**	**5.17**	**6.83**

Figs. 11–13 show the estimated abundance maps using different methods and the ground truths. It can be seen that the proposed method achieves the best results in most cases. Specifically, for the Urban data set, the proposed method produces the results closest to the ground-truth abundance maps, while the other methods cannot fully separate the similar end-members, e.g., grass and other end-members are mixed in one abundance map. For the Jasper Ridge data set, the proposed method produces the best results compared with the other methods, while the other methods have some difficulties with some specific end-members, which appears to be mixtures of other end-members. For example, VCA, PCOMM, DMaxD, and AEU confuse road and dirt, while MLM cannot fully separate water and dirt. In terms of the Samson data set, VCA, MLM, PCOMM, and DMaxD mix water with other materials, while both AEU and the proposed method produce satisfying results. The proposed method produces better results than AEU.



(a) (b) (c) (d) (e) (f) (g) (h) (i) (j) (k) (l) (m) (n) (o) (p)

Fig. 17. Visualizations of the data-guided maps and the detected outliers using different number of neighborhoods. (a)–(d) and (i)–(l) Data-guided maps of the Urban and Samson data sets. (e)–(h) and (m)–(p) Detected outliers of the corresponding data sets. From the first column to the last column: 8, 24, 48, and 80 neighborhoods.

For the Washington DC data set, because we do not have ground-truth end-members and abundance maps, we only give qualitative comparisons. Fig. 14 shows the estimated abundance maps of all the methods. From the figure, we see that VCA mixes the asphalt and the grass end-members, while PCOMM, MLM, DMaxD, and AEU cannot fully separate the water & shadow and the asphalt. In contrast, our method produces the most distinct abundance maps.

F. Ablation Study

We remove different components in our design to investigate the effectiveness of each component, including OSP, hyper-

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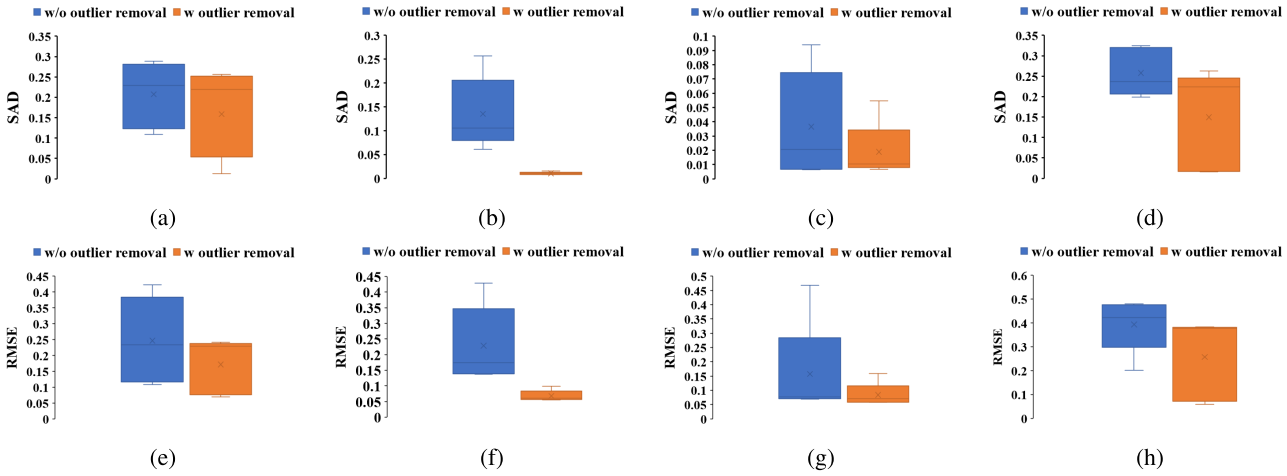


Fig. 18. End-member and abundance error statistics from the synthetic image with 1% outliers. From the first column to the last column: limestone, basalt, concrete, and asphalt.

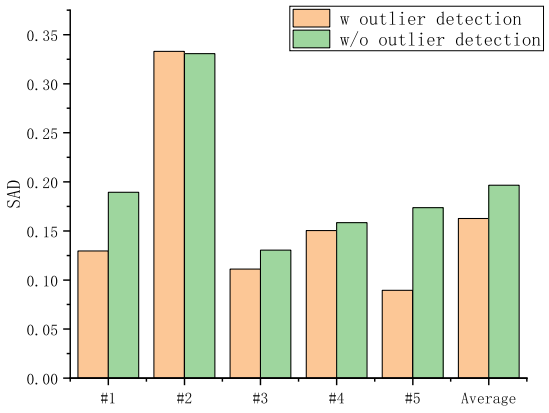


Fig. 19. Model comparisons with and without outlier detection process. MLM is used as the test unmixing algorithm, and Urban is used as the test data set.

Laplace loss, and outlier detection process. We use Urban as the test data set. Table V shows the results. The OSP is shown to be most effective that produces 57.88% better SAD compared with the baseline. The hyper-Laplacian loss and the outlier detection process also reduce the SADs by 30.5% and 16.2%, respectively. The combinations of different components could further improve the model performances. Using all components yields the best performances.

To better understand the OSP and the hyper-Laplace loss, we visualize the abundance maps without the two components. The first row of Fig. 15 shows the abundance maps without the OSP. From the figure, we see that the OSP leads to more distinct abundance maps than L_1 norm sparse prior. Since the OSP enforces not only the sparse constraint but also the orthogonality on the abundance maps, it is a stronger prior than norm-based sparse prior. The second row of Fig. 15 shows the abundance maps without the hyper-Laplace loss. Due to the presence of outliers and the differences between the reconstruction and the raw HSI, the mean square error is vulnerable to these factors; thus, it tends to mix some end-members, resulting in the ambiguous abundance maps.

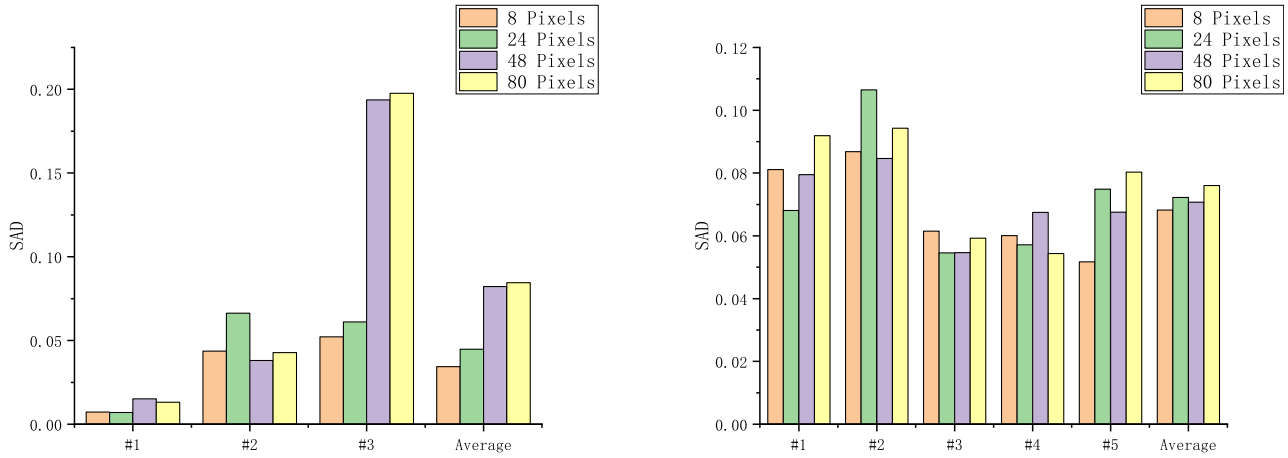
We further use the synthetic data set with 1% outliers to verify the effectiveness of the outlier detection process. Specifically, we randomly pick 1% pixels and replace them with random noise with uniform distribution. Fig. 18 shows the box plots of end-member errors. The model with the outlier detection process exhibits smaller errors than that without the outlier detection process. Fig. 16 shows the initialized end-members with and without the outlier detection process. We see that compared with the initialization without the outlier detection process, the initialized end-members with the outlier detection process are much more similar to the ground truths.

Finally, we show that the outlier detection process is a general preprocessing technique that improves the performances of the unmixing algorithm. We use MLM as the baseline unmixing algorithm. Fig. 19 shows the SAD comparisons for each end-member and the average performance, where "without outlier detection" ("with outlier detection") represents MLM without (with) the outlier detection process. Removing the outlier detection process leads to 20.77% worse SAD. The performance decline demonstrates the generality of the proposed outlier detection process.

G. Hyperparameter Sensitivity Analysis

There are eight hyperparameters in the model, including $|\mathcal{N}|$, p, σ, τ, ρ_1, ρ_2, ρ_3, and ρ_4. Among these hyperparameters, $|\mathcal{N}|$, p, τ, and ρ_1 are relatively sensitive to the model performance. p controls the heavy-tailed shape of the reconstruction error distribution, and smaller p leads to the heavier tails of the error distribution; τ controls the sensitivity of the outlier detection, and more pixels are treated as outliers with smaller τ; and ρ_1 controls the orthogonality of the abundance maps, and the larger the ρ_1, the stronger the orthogonality of the abundance maps. Table VI shows the mean SADs with respect to different p, τ, ρ_1, ρ_2, ρ_3, and ρ_4. We use Urban as the test data set. We see that p, τ, and ρ_1, which are too small or too large, lead to performance degradation, while ρ_2, ρ_3, and ρ_4 are less sensitive to the model performance. We, therefore,

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(a) (b)

Fig. 20. Mean SAD with respect to different number of neighborhoods on two data sets. (a) Samson data set. (b) Urban data set.

TABLE VI

MEAN SADS ($\times 10^{-2}$) WITH RESPECT TO DIFFERENT p, τ, ρ_1, ρ_2, ρ_3, AND ρ_4

p	0.5	0.6	0.7	0.8	0.9
SAD	8.43	7.89	**6.83**	7.98	8.31
τ	0	0.3	0.5	0.7	0.9
SAD	9.4	7.54	**6.83**	7.93	8.35
ρ_1	0.1	0.5	1	1.5	2
SAD	9.87	**6.83**	7.02	7.51	8.35
ρ_2	0.1	0.5	1	1.5	2
SAD	7.01	6.95	**6.83**	7.00	7.02
ρ_3	1	5	10	15	20
SAD	7.00	6.91	**6.83**	6.96	6.97
ρ_4	1	5	10	15	20
SAD	6.99	6.97	**6.83**	7.01	6.98

use $p = 0.7$, $\tau = 0.5$, $\rho_1 = 0.5$, $\rho_2 = 1$, and $\rho_3 = \rho_4 = 10$ throughout this article.

In terms of $|\mathcal{N}|$, theoretically, when one end-member occupied a large area, using a large number of neighborhoods may be more suitable for detecting outliers; however, when multiple end-members are scattered and mixed in one region, using a small number is a better choice. We compare the unmixing performances using a different number of neighborhoods on two data sets. Specifically, we use 8 (3×3 window size), 24 (5×5 window size), 48 (7×7 window size), and 80 (9×9 window size) as the number of neighborhood pixels. Urban and Samson are used as the test data sets. Fig. 20 shows the results. The proposed method with eight numbers of neighborhoods performs the best results on both data sets. Fig. 17 further shows visualizations of the data-guided maps and the detected outliers using a different number of neighborhoods. We see that as the number of neighborhoods increases, the data-guided maps become a blur, and more pixels are treated as outliers. Due to the complexities of end-member distributions in the real scenario, using large numbers may mistake some end-members for outliers. We, therefore, use eight throughout this article.

TABLE VII

COMPUTATION TIME (SECONDS) COMPARISONS ON URBAN DATA SET

VCA	PCOMM	MLM	AEU (train)	AEU (test)	DmaxD	Proposed (train)	Proposed (test)
0.26	77.19	371.74	45.8	0.05	3.79	45.2	0.05

H. Computing Time Comparison

We compare the computing time of all test methods, and the results are shown in Table VII. All the experiments are conducted on the same PC with an Intel i7 CPU, 8-GB memory and one TAITIAN Xp graphic card. The proposed method is trained using GPU due to its friendly GPU support, while the compared methods are tested using CPU. We see that the autoencoder-based methods take a relatively long time to train. However, once the model is well trained, it takes much less time to test the data set compared with the other state-of-the-art methods.

IV. CONCLUSION AND FUTURE WORK

In this article, we proposed a new model for spectral-unmixing based on autoencoder networks with three novel components. First, we proposed a new sparse prior, called OSP, to explore orthogonality between abundances. The key observation was that the abundance maps of different end-members are close to orthogonal because only one or two end-members exist for every pixel. We also showed that the reconstruction error has a heavy-tailed shape, which does not conform to the conventionally used Gaussian distribution. Based on this fact, we used the hyper-Laplace distribution to model the reconstruction error and proposed the hyper-Laplacian loss. Finally, we proposed a data-driven approach to detect outliers from the raw HSI to improve the performances of end-member initializations. Extensive experiments on both synthetic and real-world data sets showed that the proposed method significantly and consistently outperforms the compared state-of-the-art methods, with up to more than 50% improvements.

The proposed method is based on LMMs. However, as mentioned in the introduction, the NLMM usually better illustrates

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the physical properties of mixing pixels. How to properly incorporate the NLMM into the proposed method is an interesting research direction. Besides, the proposed OSP does not hold when multiple end-members are mixed. Recognizing these pixels and excluding them from the proposed prior may further improve model performances. We leave these two problems for future work.

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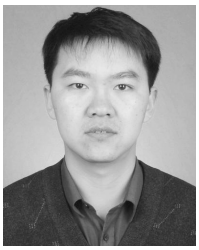
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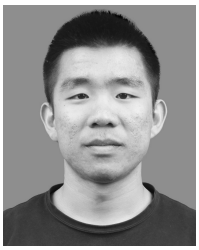
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